COMPUTATIONAL METHODS FOR PRECISION SIGNAL SYNTHESIS IN QUANTUM SENSING SUBSYSTEMS

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Introduction

- Cold atom interferometry (CAI) represents a frontier in precision measurement technology, enabling direct observation of gravitational acceleration, rotations, and fundamental constants with quantum-limited accuracy.

- At the heart of a cold atom interferometer, typically involve a combination of Raman or Bragg laser pulses that coherently split, redirect, and recombine the atomic wave packets.

- Therefore, precise control of multiple laser beams, frequency references, and timing signals is an essential engineering challenge in such experiments.

- To address this, a custom-built embedded control system has been developed, designed around an STM32 microcontroller, a Direct Digital Synthesizer (DDS), and an FPGA-based Phase- Comparator.

Quantum gravimeter subsystem prepared by TCG CREST

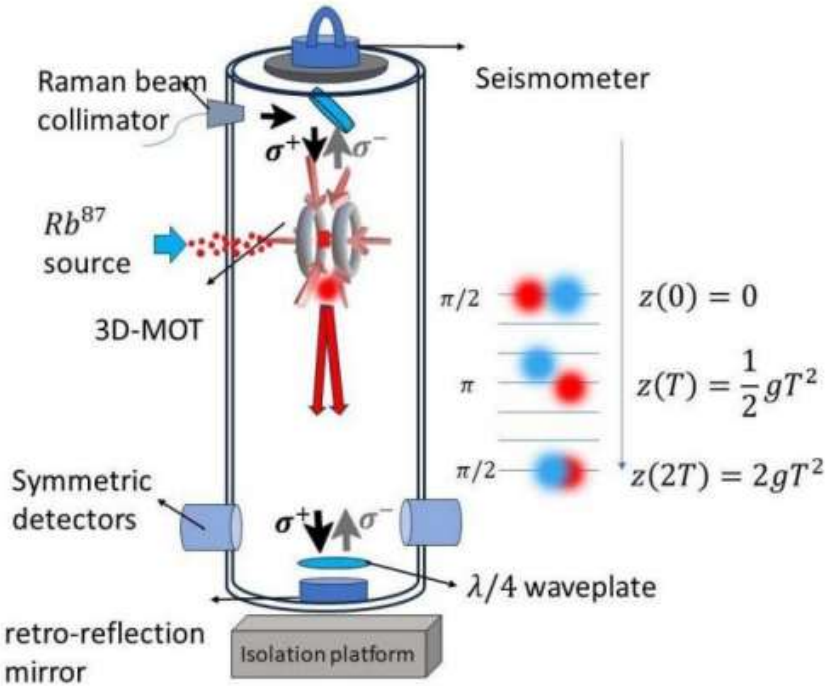


Fig. 1. Quantum Gravimetry Subsystem Principle[1]

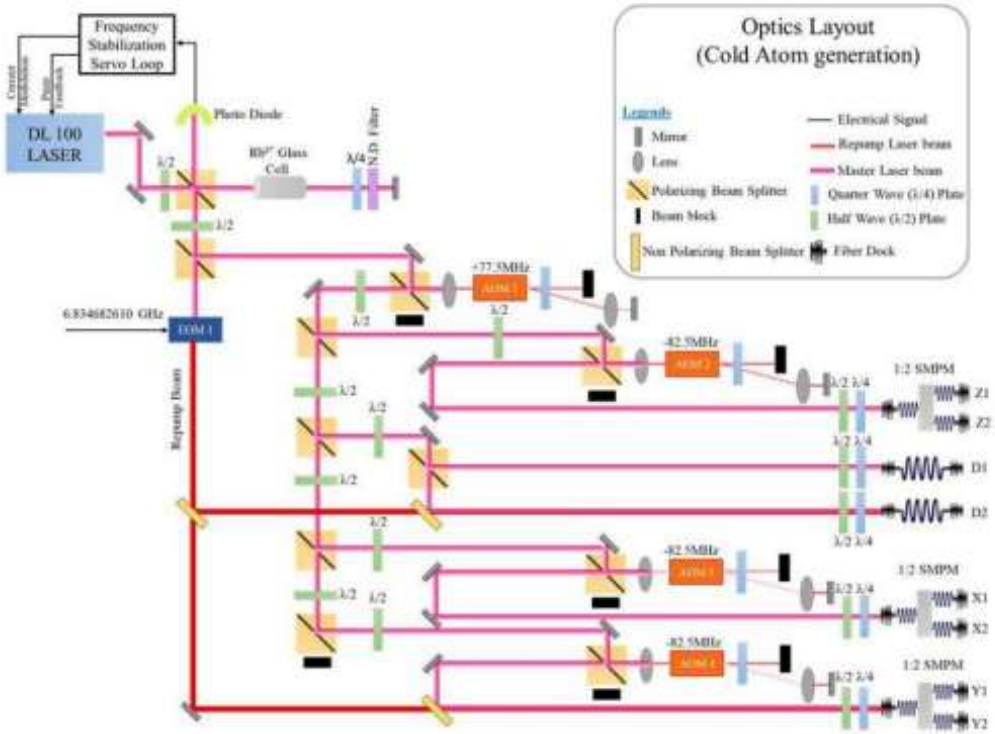
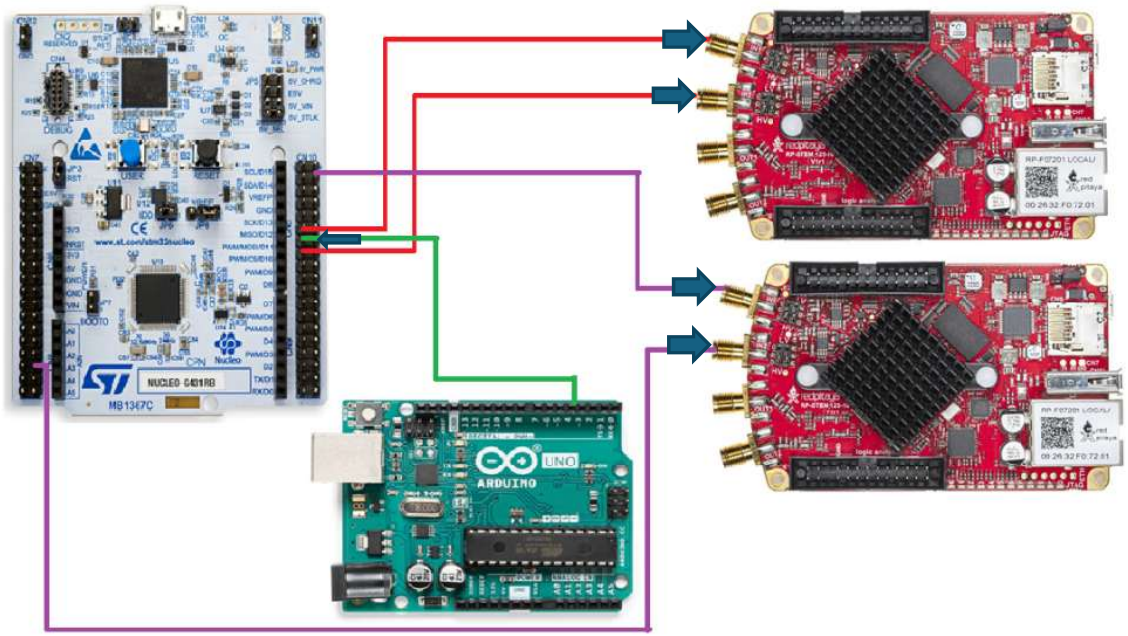


Fig. 2. Optics Layout for Cooling and Detection[1]

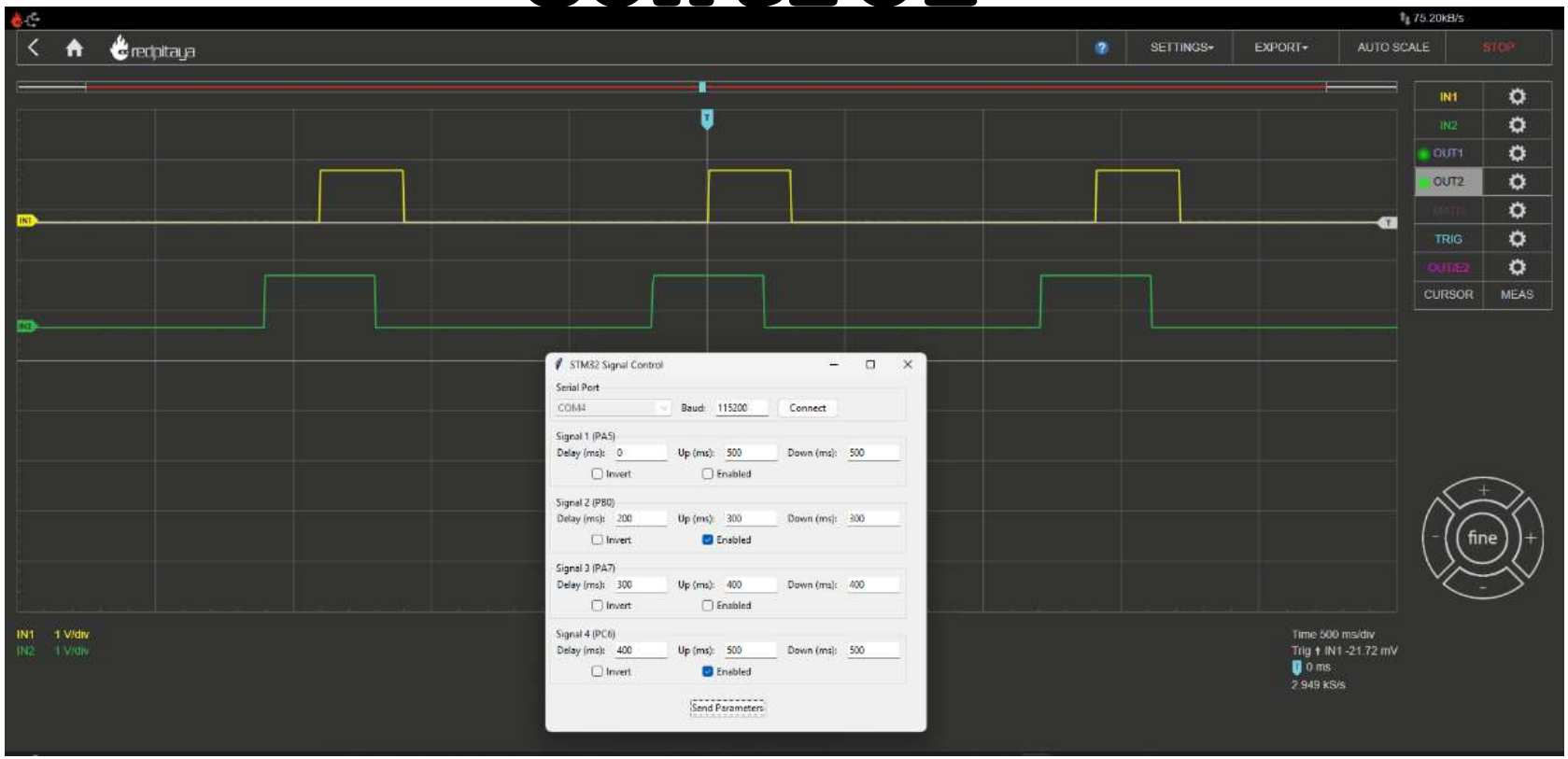
Signal generation using STM32 G431RB for multiple pulse control



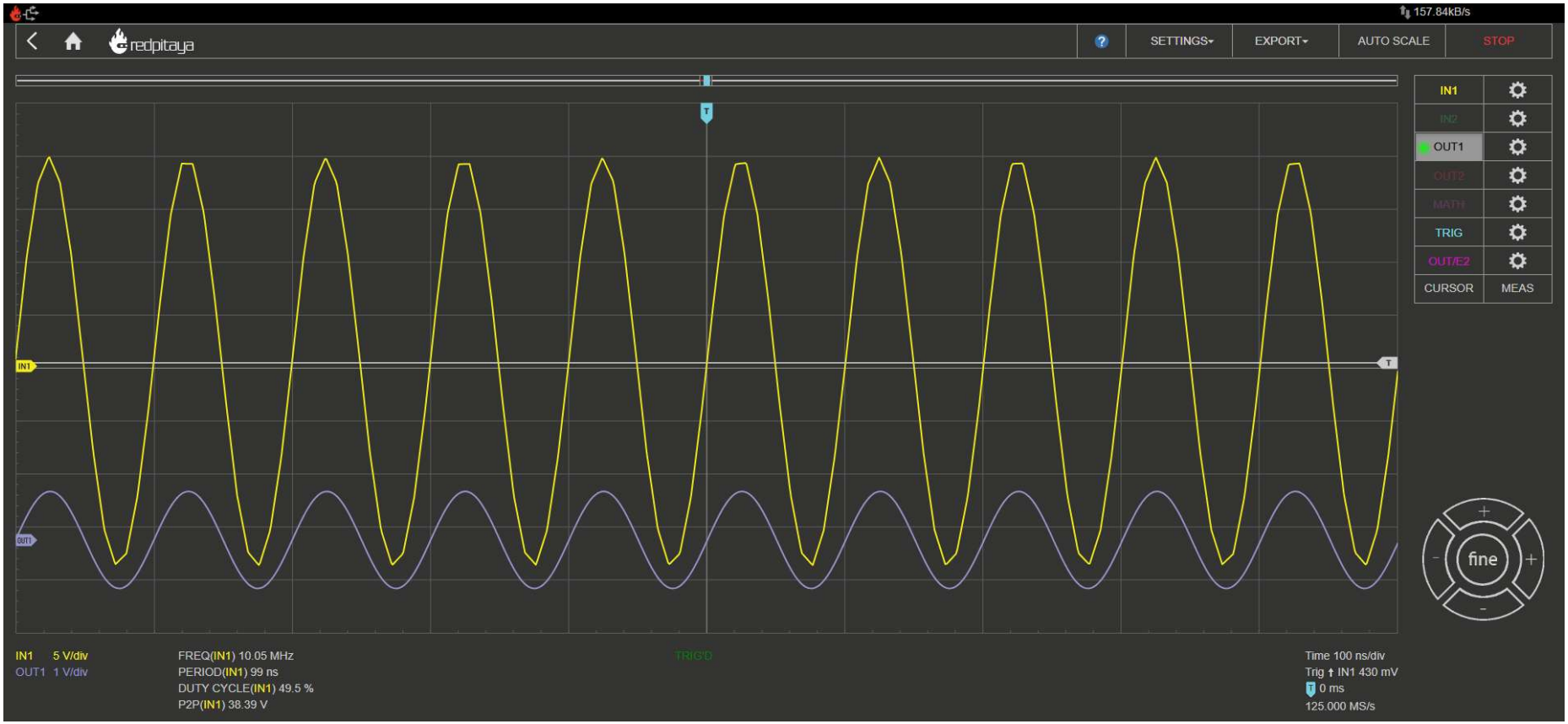
Signal generation using STM32 G431RB for multiple pulse control

The STM32G4 Nucleo-64 boards based on the MB1367 reference board (NUCLEO-G431RB, NUCLEO-G474RE, NUCLEO-G491RE) provide an affordable and flexible way for users to try out new concepts and build prototypes with the STM32G4 series microcontrollers,
choosing from the various combinations of performance, power consumption, and features. Our system consists of an STM32G431RB microcontroller responsible for generating four independent digital control signals used to operate beam shutters. Each signal can be programmed with specific parameters including delay, pulse width, polarity, and enable status. The control signals are synchronized with an external trigger pulse generated by an Arduino board. The entire setup is tested and validated using a Red Pitaya, which provides real-time visualization and waveform analysis.

Signal generation using STM32 G431RB for multiple pulse control

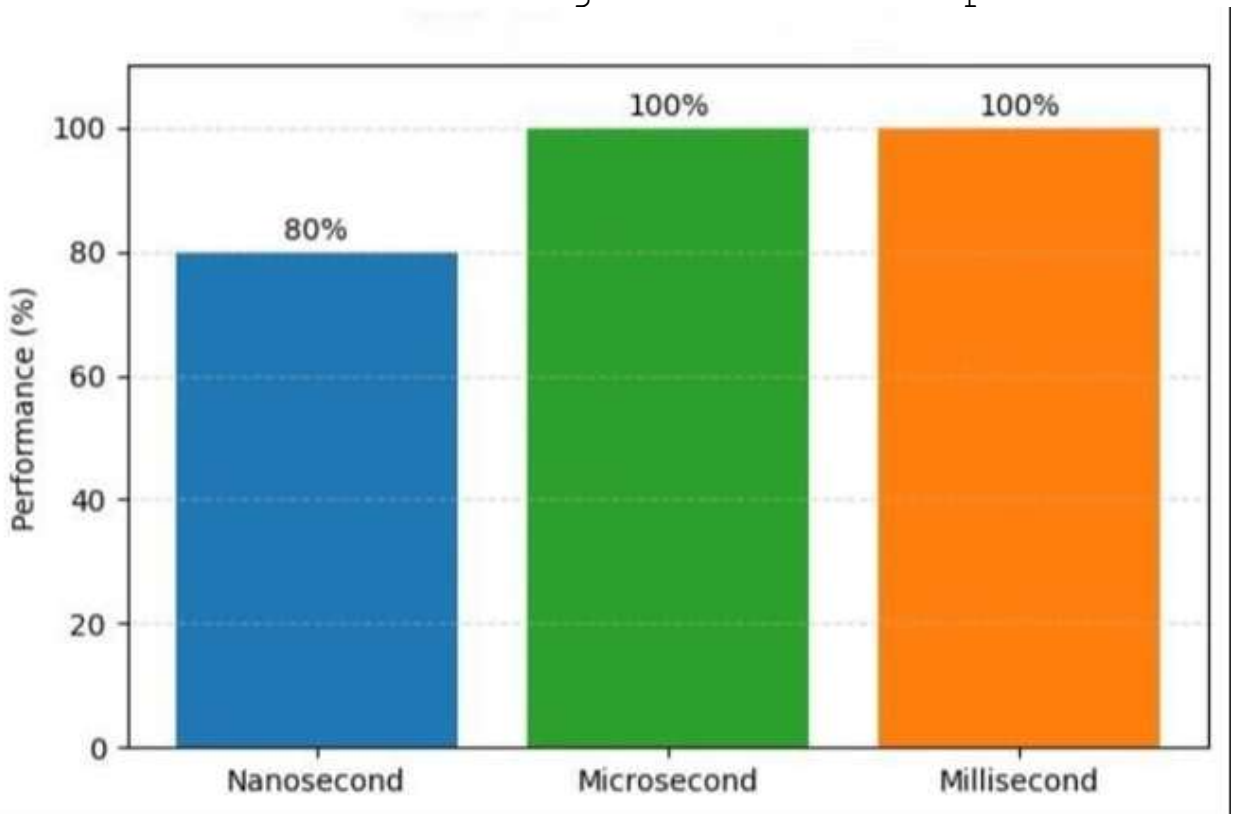


Waveform triggered by DDS

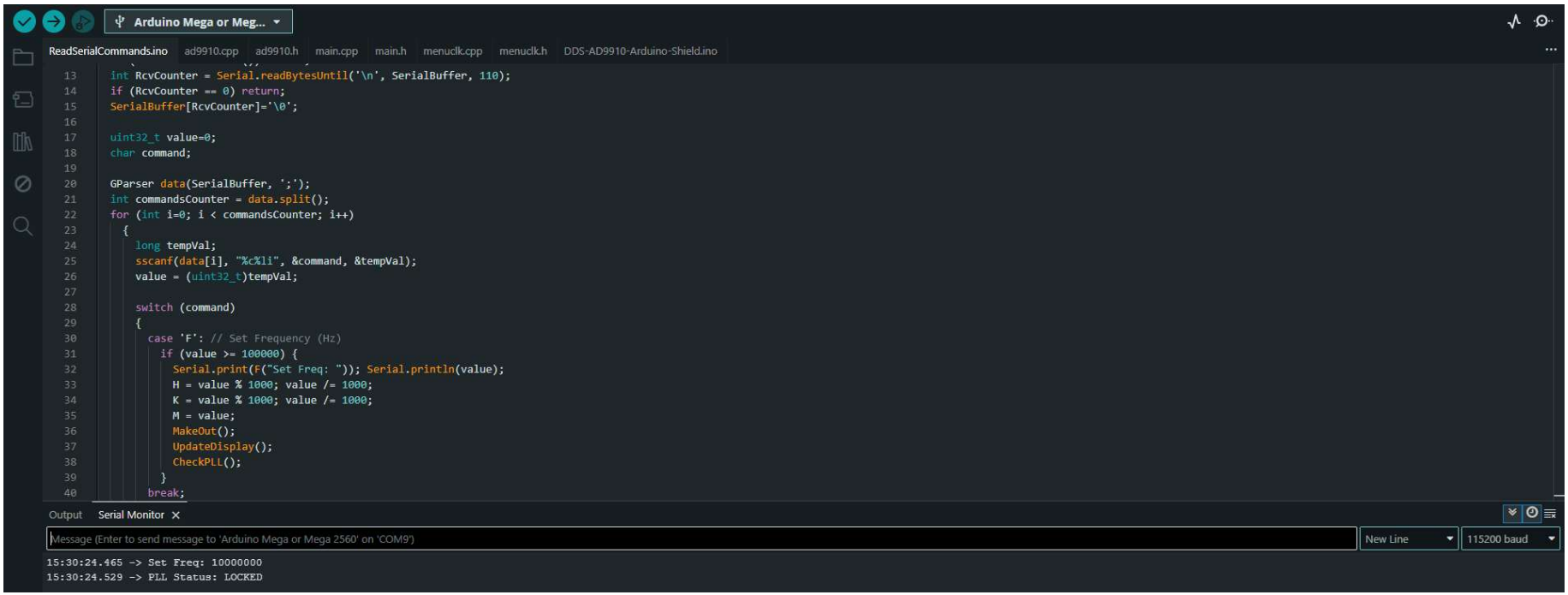


Result Accuracy

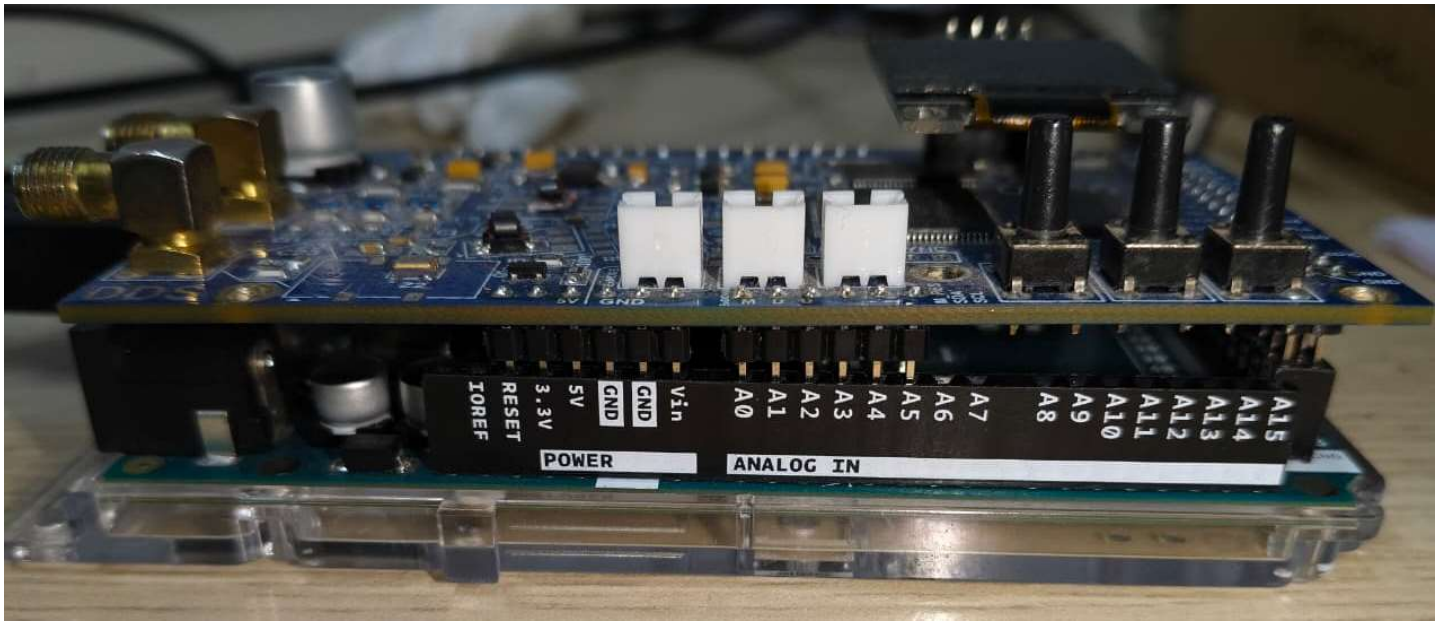
Timing Precision Graph



DDS code for PLL locking on FPGA



Signal generation using DDS to simulate laser

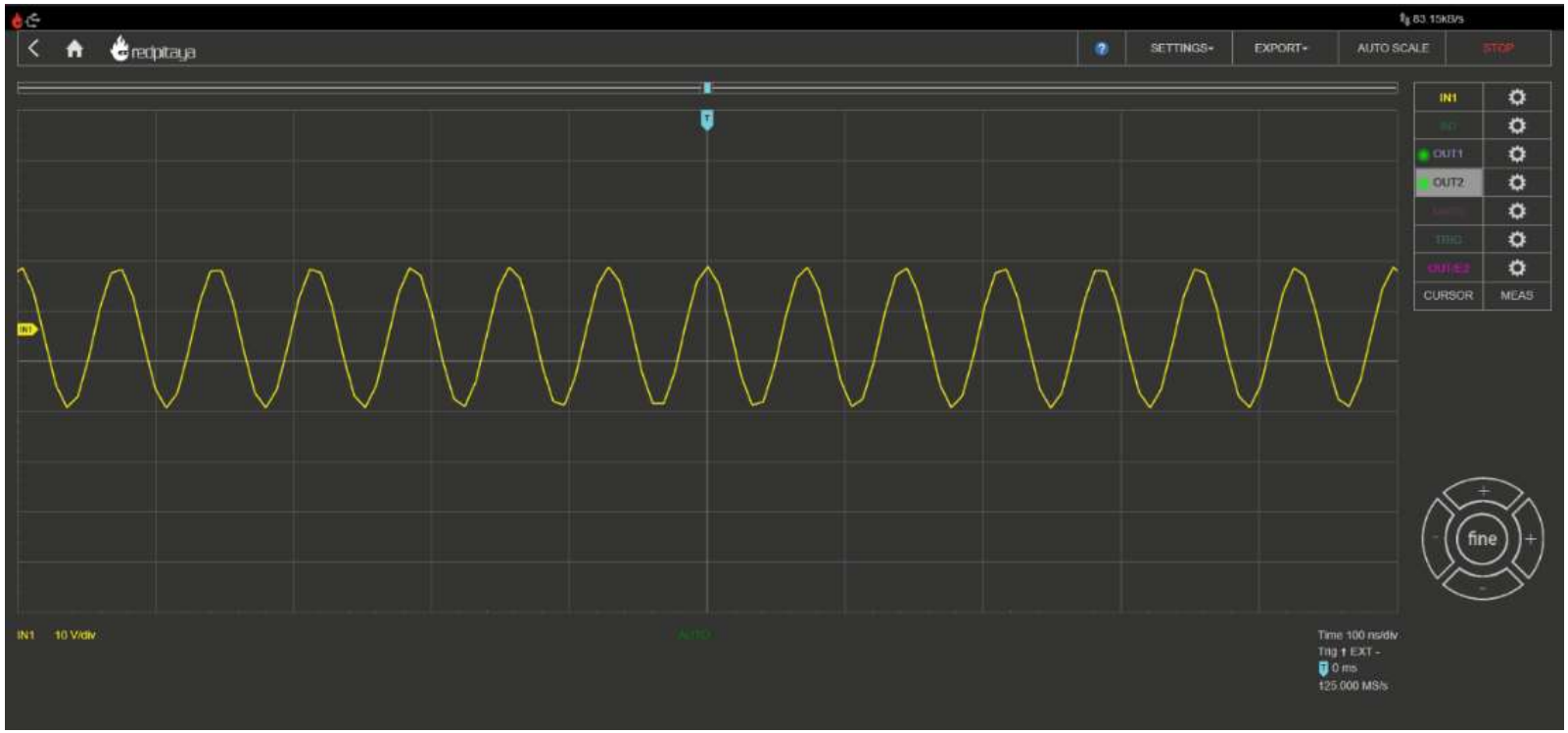


Signal generation using DDS to simulate laser

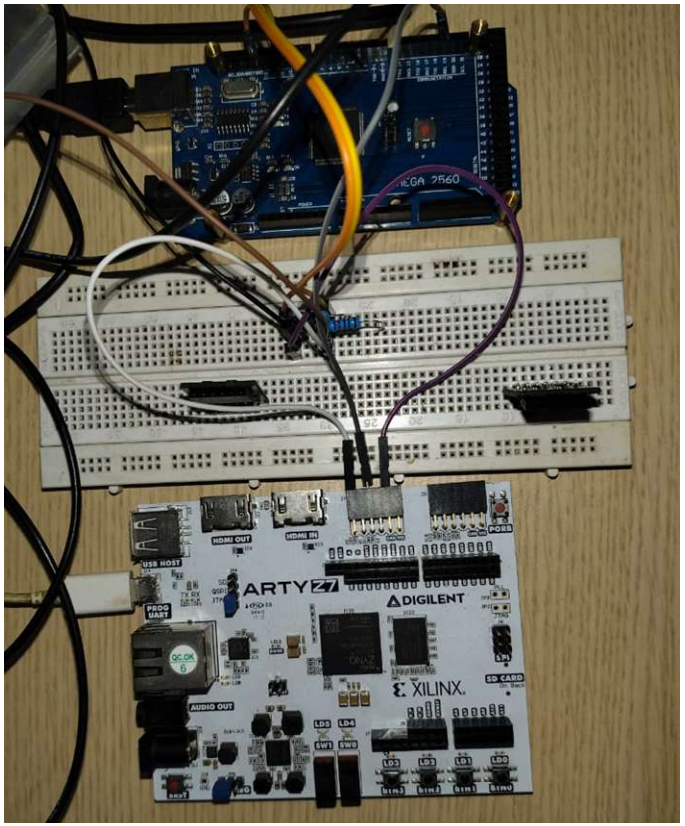
In quantum gravimetry experiments based on cold atom interferometry, precisely controlled laser frequencies are essential for manipulating atomic states through Raman or Bragg transitions. The Direct Digital Synthesizer (DDS) offers an excellent solution by providing fine frequency, phase, and amplitude control with high spectral purity and fast switching capability.

In this setup, the Arduino Mega 2560 is used as the control interface for the AD9910 DDS, enabling digital programming of frequency tuning words, phase offsets, and profile registers via the SPI communication protocol.

Signal generation using DDS to simulate laser



Signal generation using FPGA to make a Phase Comparator

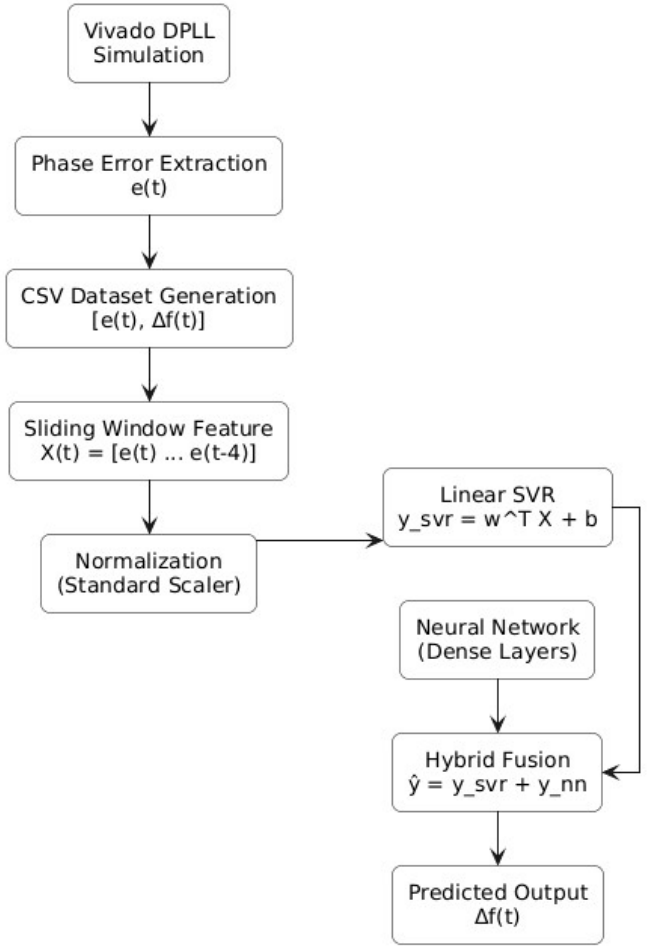


Signal generation using FPGA to make a Phase Comparator

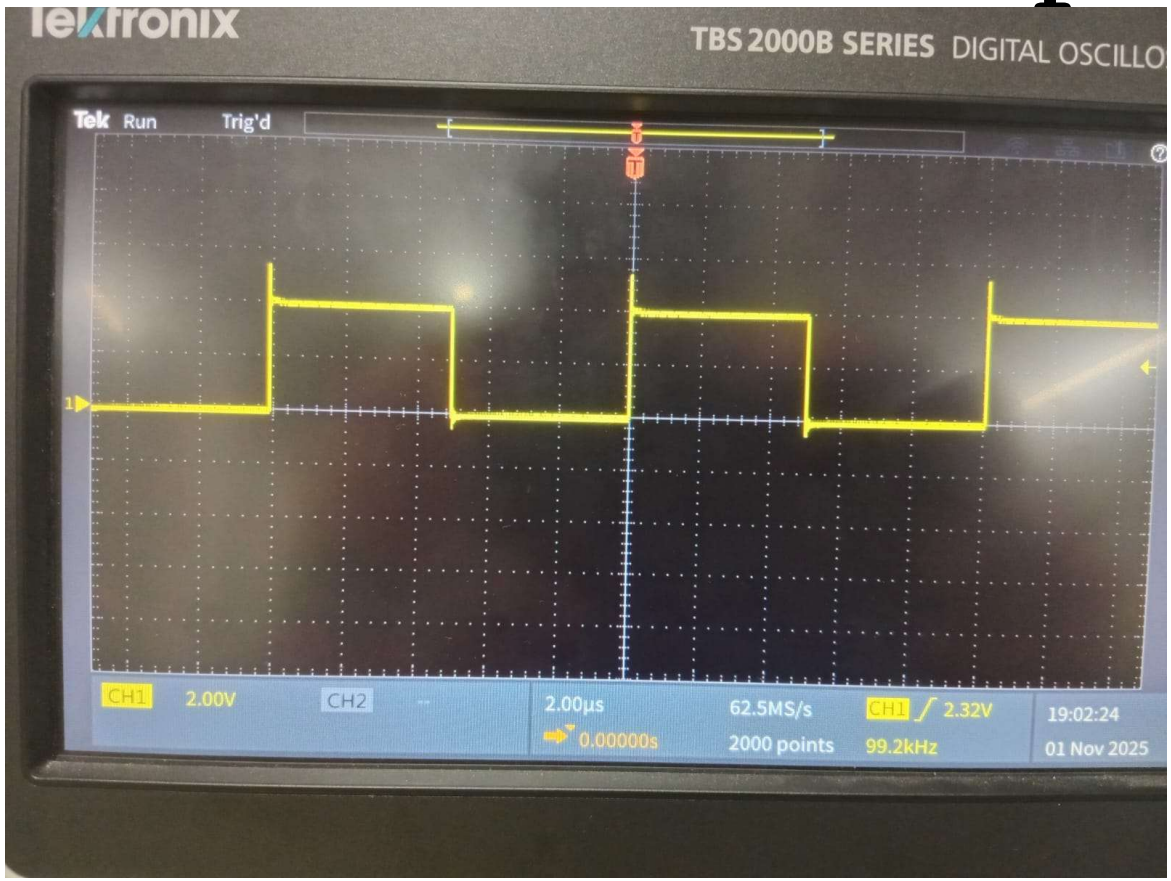
Quantum gravimetry relies on the precise control and synchronization of electromagnetic signals that drive laser-based atom interferometers. Accurate phase coherence between reference and experimental signals is critical for detecting the minute phase shifts induced by gravitational acceleration on cold atoms. To achieve this level of control, stable frequency generation and precise phase comparison hardware are essential.

In this project, a Field-Programmable Gate Array (FPGA) is employed to generate a reference frequency derived from its internal 125 MHz system clock.

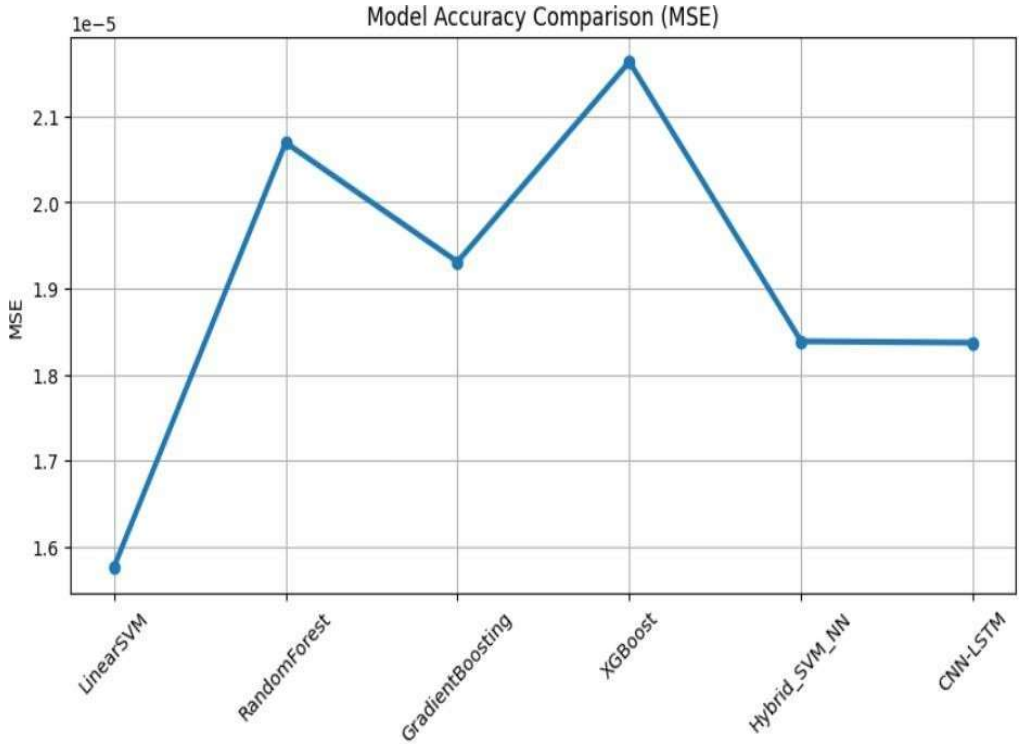
Proposed Methodology



Signal generation using FPGA to make a Phase Comparator



Results & Discussion



Model	MSE
Linear SVM	1.5798×10^{-5}
Random Forest	2.0671×10^{-5}
Gradient Boosting	1.9329×10^{-5}
XGBoost	2.1632×10^{-5}
Hybrid SVM–NN	1.8620×10^{-5}
CNN–LSTM	1.8541×10^{-5}

TABLE I

MSE COMPARISON ACROSS MODELS

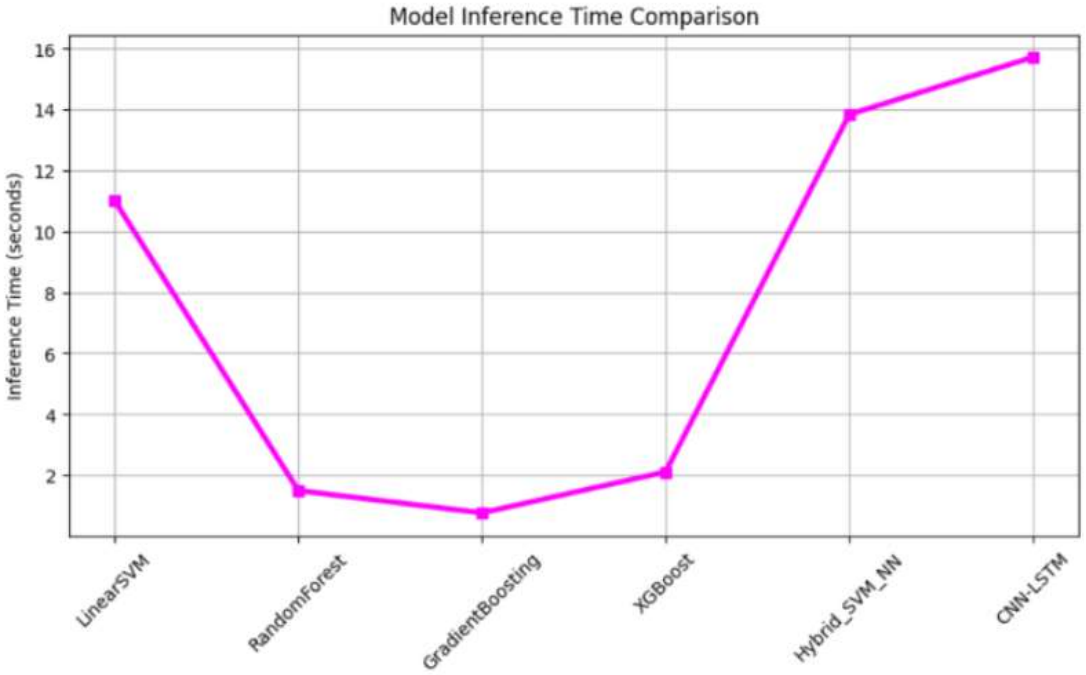


Fig 6.12: Inference Time Comparison between different Models

Model	Inference Time (s)
Linear SVM	6.2381
Random Forest	0.5466
Gradient Boosting	0.3169
XGBoost	0.3180
Hybrid SVM–NN	8.0494
CNN–LSTM	15.6110

Table 6.2. Inference time of Different ML Models

Future Work

In future we want to :

- Synchronize all these systems together to ease up the quantum gravimetry based experiments.
- We will also create a AI Assisted FeedBack Loop integrated with the Arty Z7 FPGA.

Thank You

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&
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